Email Thread Summarization: Leveraging Event and Trigger Extraction

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Abstract

To address "inbox fatigue," we evaluate a DistilBART-based email thread summarization pipeline using event and trigger extraction on a subset of Enron: MailEX. Results indicate that while ROUGE and BERTScore metrics show surface-level similarities, trigger-based anchoring is significantly more precise: enhances role assignment and logistical accuracy. This demonstrates that anchoring improves factual precision and generalizes across different reference summaries (Gemini and OpenAI).

1 Introduction

Email thread summarization is highly effective for addressing 'inbox fatigue', however, generated summaries often lack precision. Therefore, this study specifically aims to evaluate how the extraction of triggers (words initiating an action) and events (the type of action) can assist in fine-tuning models to produce more accurate thread summaries.

2 State of the art

Email summarization has evolved from traditional information extraction approaches to modern neural and transformer-based models, yet it remains a challenging task due to the unique characteristics of email communication. One major challenge is the multi-party and threaded structure of email conversations, which requires models to track context across multiple messages and speakers to preserve the flow of information (Zhang et al., 2021). Another challenge is redundancy from replies that include quotes from previous emails, which can lead models to generate repetitive summaries (Zhang et al., 2021) (Khetarpaul et al., 2025). A third difficulty is the informal and domain-specific language of emails, which often necessitates fine-tuning for accurate summarization (Srivastava et al., 2023).

To benchmark progress, several datasets have been developed. Among the most popular, EMAIL-SUM provides human-annotated abstractive summaries derived from the Enron email corpus (Zhang et al., 2021), while MailEx, also derived from Enron, focuses on extracting information such as events and triggers (Srivastava et al., 2023). More recently, large language models (LLMs) such as GPT-3, Google Bard, and LLaMA2 have been explored for email-thread summarization, demonstrating improved fluency and context understanding compared to traditional models (Khetarpaul et al., 2025). GPT-3, for instance, produces summaries closer to human-written references, although it can struggle with long-range dependencies and occasional verbosity. Bard and LLaMA2 show complementary strengths, but evaluation across multiple threads highlights that human judgment still surpasses these models.

Despite these advances, prior research has largely treated event extraction and summarization as separate tasks. In this work, we propose exploring the combination of MailEx's extracted events and triggers with a pretrained summarization model, to evaluate whether integrating structured information can enhance the informativeness and context-awareness of generated summaries, using ROUGE (Lin, 2004) and BERTScore (Zhang et al., 2020) for performance assessment.

3 Methodology

3.1 Data Processing and Knowledge Distillation

We utilize the MailEx (Srivastava et al., 2023) dataset, a subset of Enron (Corp. and Cohen, 2015) containing ∼1.4K threads and ∼9K event annotations. Each thread includes structured JSON meta-

data identifying event types, BIO-tagged triggers, and meta-semantic roles. Because MailEx lacks narrative summaries, we employed knowledge distillation to generate "silver-label" targets. We prompted the OpenAI API (GPT-4.1 nano, (OpenAI, 2023)) and the Gemini API (Gemini 2.5 flash lite, (Team and Google, 2023)) with raw threads and their corresponding triggers/arguments to produce professional, grounded summaries. This few-shot approach ensures the labels remain faithful to the annotated events. The dataset was partitioned into 1,200 training and 150 test instances.

3.2 Model Selection: DistilBART

While large language models (e.g., Llama 3B (Dubey et al., 2024)) offer high performance, their memory requirements exceeded our hardware constraints (NVIDIA T4 16GB). We instead selected DistilBART-CNN-12-6 (Shleifer and Lewis, 2020), a Seq2Seq model pre-trained on the CNN/DailyMail corpus. DistilBART retains 97% of BART-Large's (Lewis et al., 2019) performance while significantly reducing parameter count and latency, making it a high-performing student model for domain-specific distillation.

3.3 Finetuning via Feature Priming

The model was finetuned for 5 epochs using a Feature-Priming strategy. Rather than providing unstructured text, the input was formatted as: summarize: [Triggers/Arguments] THREAD: [Raw Text]. This anchoring technique utilizes the model's cross-attention mechanism to prioritize event-heavy segments over conversational noise. We utilized the Seq2SeqTrainer with a learning rate of 5×10^{-5} and FP16 precision. To minimize hallucinations during inference, we employed Beam Search ($k = 5$) and a trigram repetition penalty.

4 Results

We evaluate our summarization models using ROUGE (ROUGE-1, ROUGE-2, ROUGE-L, and ROUGE-Lsum) to assess lexical overlap and BERTScore (F1) to measure semantic similarity between generated and reference summaries. Table 1 presents the results when evaluating against OpenAI-generated reference summaries. Both configurations (models trained with triggers and without triggers) achieve scores within commonly accepted good performance ranges. The ROUGE scores indicate comparable levels of content coverage across both settings, and BERTScore values (0.8326 with triggers vs. 0.8342 without triggers) suggest similar degrees of semantic alignment with the reference summaries.

A parallel evaluation was conducted using Gemini-generated reference summaries. As shown in Table 2, the overall pattern remains consistent. ROUGE and BERTScore values again fall within the good performance ranges and differences between the with-trigger and no-trigger configurations are marginal.

Metric	With triggers	No triggers	Range
ROUGE-1	0.4741	0.4887	0.4 – 0.6
ROUGE-2	0.2381	0.2602	0.2 – 0.4
ROUGE-L	0.3382	0.3581	0.3 – 0.5
ROUGE-Lsum	0.3385	0.3583	0.3 – 0.5
BERTScore F1	0.8326	0.8342	0.8 – 0.9

Table 1: Comparision of ROUGE and BERTScore (based on their commonly accepted good performance range) for generated summaries given triggers vs. no triggers evaluated against OpenAI reference summaries.

Metric	With triggers	No triggers	Range
ROUGE-1	0.5615	0.5457	0.4 – 0.6
ROUGE-2	0.3186	0.3143	0.2 – 0.4
ROUGE-L	0.4150	0.4177	0.3 – 0.5
ROUGE-Lsum	0.4147	0.4174	0.3 – 0.5
BERTScore F1	0.8689	0.8649	0.8 – 0.9

Table 2: Comparison of ROUGE and BERTScore (based on their commonly accepted good performance range) for generated summaries given triggers vs. no triggers evaluated against Gemini reference summaries.

However, a manual comparison of selected generated summaries reveals qualitative differences between the two configurations. When triggers are provided, the model often produces higher-quality summaries that more accurately capture and preserve the meaning and logical structure of the email threads. Although both versions generally cover similar content, summaries generated without triggers sometimes alter the intended meaning or produce statements that lack coherence. Figure 1 illustrates such a case: without triggers, the model produces the logically inconsistent statement "the old Enron building is the new one or the old one," whereas the trigger-based version preserves the intended semantics ("the building was the old one or the new one"). Additionally, the no-trigger summary presents the events in reversed order, leading to reduced contextual clarity.

Improvements in accuracy and logistical detail extraction are also observable when triggers are included. As shown in Figure 2, the version without

triggers fails to clearly identify the person responsible for organizing the initiative and instead begins with a passive formulation ("The email thread discusses. . . "). In contrast, the trigger-based summary explicitly identifies Rahil as the organizer of the session. Moreover, with triggers the model correctly extracts the time for the meeting ("scheduled for tomorrow"), whereas without triggers the model simply states that the meeting is scheduled, without specifying the time.

Thread: Email 0: Please come by 5C2 at your convenience tomorrow as UBS require you to complete missing information on your application form . Kind regards , Karen Buckley Human Resources Tel : 713 345 4667
Email 1: Is this the old enron building or the new one ? Eric Bass
Email 2: old one . thanks ! .

Summary (given triggers): Karen Buckley asked recipients to come by 5C2 at their convenience tomorrow to complete missing information on their application form. Eric Bass asked if the building was the old one or the new one.

Summary (no triggers): Eric Bass asked if the old Enron building is the new one or the old one. Karen Buckley asked Eric to come by 5C2 at his convenience tomorrow to complete missing information on his application form.

Figure 1: Example of generated summaries given triggers vs. no triggers from a DistilBART model trained on OpenAI reference summaries.

Thread: Email 0: Everyone , As we prepare to re-launch the website , we 're planning a simulation session for our traders and originators for next week (tentatively Monday) . This session will be similar to the ones we had before the initial launch of EnronOnline . In brief , the goals are : Test the integrity/functionality of the system and related processes , Get everyone re-acclimated with the system and processes (traders , originators , credit , operations , help desk , et al) , and Show how the trading system and our processes work to the UBS staff visiting next week . In order for the simulation to be successful , we will need to ensure that the simulation environment is exactly like the " live " trading environment . We will need to coordinate with each one of you about what needs to get done to make this possible . I will be sending out an invite for a meeting tomorrow morning . Please make sure you attend to discuss what needs to get done on your end (setting up trader/counterparty profiles , credit headroom , etc .) , as this simulation has been requested for early next week . If you have prior commitments , please appoint someone from your group to fill in for you . Feel free to call me with any questions . Regards , Rahil 3-3206
Email 1: FYI - Meeting is at 0900 (CST) in ECS06716 - Phone number : 713.584.1109

Summary (given triggers): Rahil is planning a simulation session for UBS traders and originators for next week, tentatively Monday. The session aims to test the integrity of the system and processes, and to show UBS staff how the trading system works to them. Participants need to coordinate with each other about what needs to get done. A meeting is scheduled for tomorrow at 09:00 CST in ECS06716 with phone number 713584.1109.

Summary (no triggers): The email thread discusses a simulation session for traders and originators scheduled for next week, tentatively Monday, to prepare for re-launch the EnronOnline website. Participants are asked to coordinate on their end, setting up trader/counterparty profiles, credit headroom, and to ensure the simulation environment is exactly like the "live" trading environment. The meeting is scheduled for tomorrow.

Figure 2: Example of generated summaries given triggers vs. no triggers from a DistilBART model trained on OpenAI reference summaries.

5 Conclusion

To conclude, our findings indicate that metrics such as ROUGE and BERTScore do not truly reflect textual accuracy, particularly in the context of summarization. Implementing triggers has proven to yield more accurate results across various reference summaries (from Gemini and OpenAI). This demonstrates the positive effect of trigger-based anchoring in accuracy.

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*Marina Hornero Merino : State of the art; Data exploration; Pipeline design;
*Krisa Carka : Extraction of the triggers and prompt design; Evaluation with ROUGE & BERTScore; Results analysis;
*Youssef EL Otmani : Extraction of the triggers and prompt design; Model Distillbart without triggers; Results analysis;
*Cristian-Ioan Bratu : Pipeline design; Generation of OpenAI and Gemini summaries; Model Distillbart with triggers;

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